SOC Analyst Investigation Report

1. Introduction

This report presents a simulated Security Operations Center (SOC) investigation focusing on multi-protocol traffic analysis. The objective is to identify normal and abnormal network behavior using ICMP, DNS, and HTTP protocols. The investigation replicates real-world detection and analysis techniques used by professional SOC analysts.

2. Protocols Analyzed

ICMP: Used for network diagnostics. High-frequency usage may indicate scanning or flood attacks.

DNS: Resolves domain names to IP addresses. Abnormal queries may indicate malware or DGA activity.

HTTP: Unencrypted communication protocol. Exposes sensitive data in plaintext.

3. Methodology

Traffic was generated using ping for ICMP, nslookup for DNS queries, and browser activity for HTTP. Wireshark was used to capture and analyze traffic. Filters were applied to isolate specific behaviors and identify anomalies.

4. ICMP Analysis

Continuous ICMP traffic was observed with sequential packet patterns and minimal delay.

Key Insight: This behavior indicates automated traffic and resembles ICMP flood attack patterns.

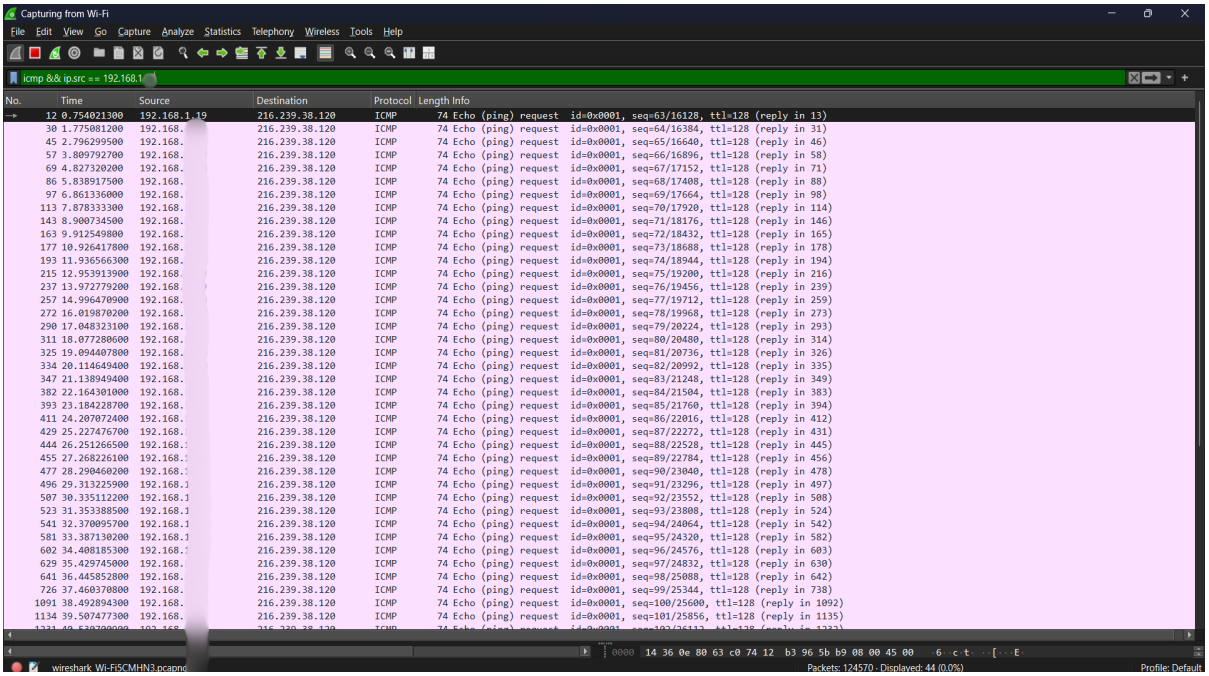

Figure 1: Continuous ICMP Traffic

5. DNS Analysis – Suspicious Queries

Multiple queries to random and non-existent domains were observed.

Strong Indicator: Random domain queries suggest Domain Generation Algorithm (DGA) behavior.

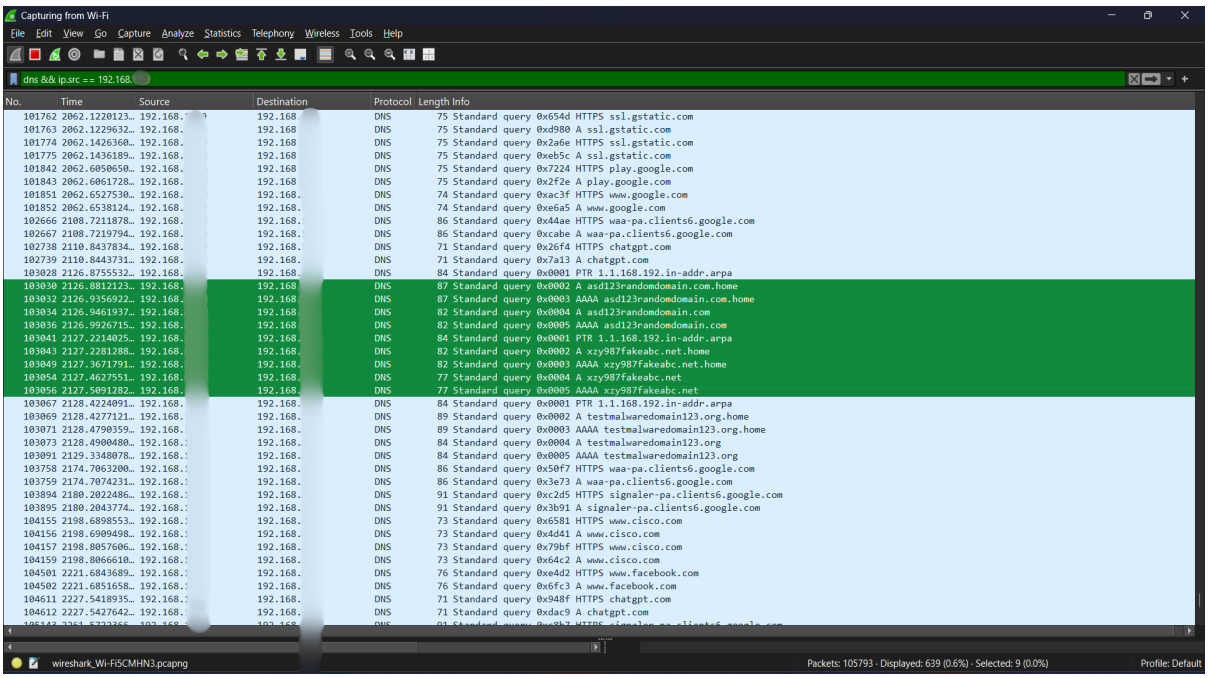

Figure 2: Suspicious DNS Queries

6. DNS Analysis – NXDOMAIN Responses

Failed DNS responses (NXDOMAIN) confirm that queried domains do not exist.

Critical Insight: High NXDOMAIN rates are strong indicators of automated or malicious activity.

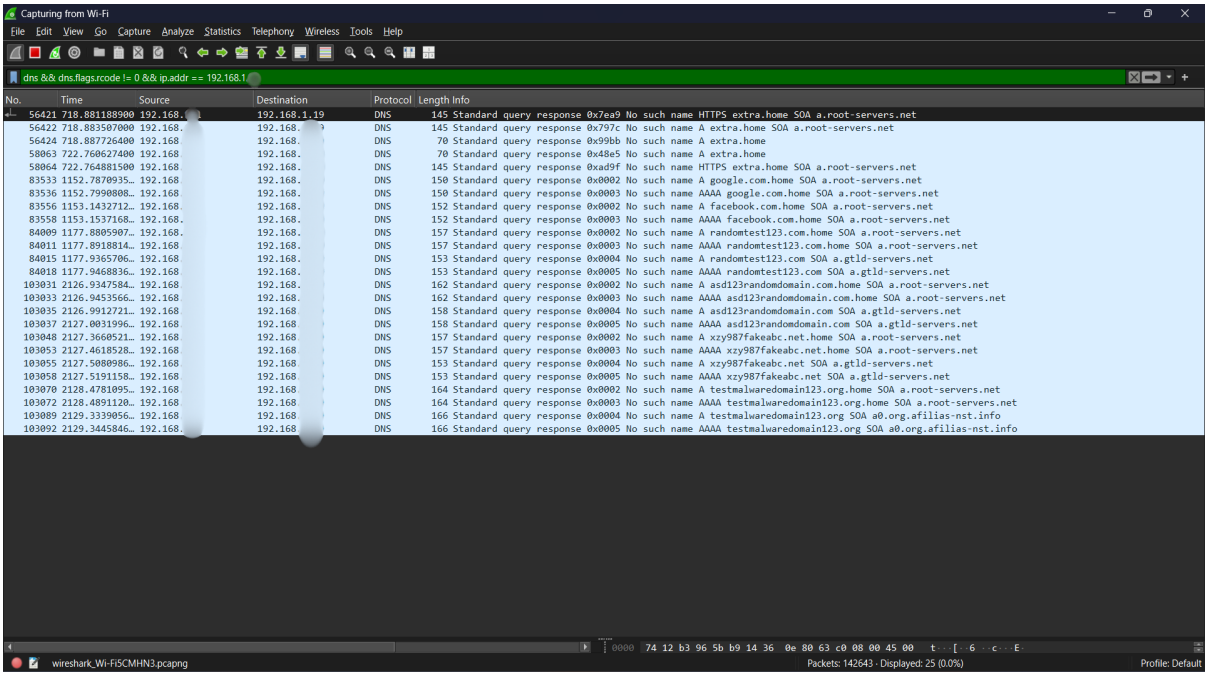

Figure 3: DNS Failed Responses

7. Simulation Evidence

Manual nslookup commands were executed to simulate malicious DNS behavior.

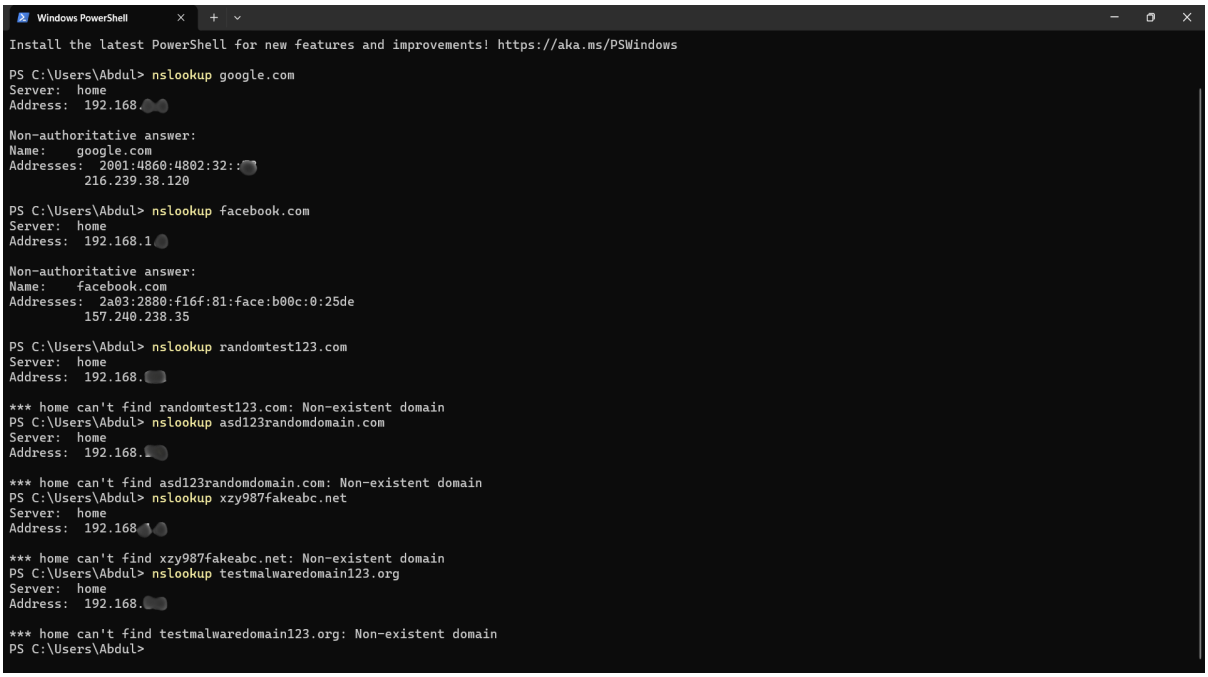

Figure 4: DNS Simulation via nslookup

8. HTTP Analysis

HTTP traffic revealed unencrypted GET and POST requests.

Security Risk: Data transmitted in plaintext can be intercepted in Man-in-the-Middle (MITM) attacks.

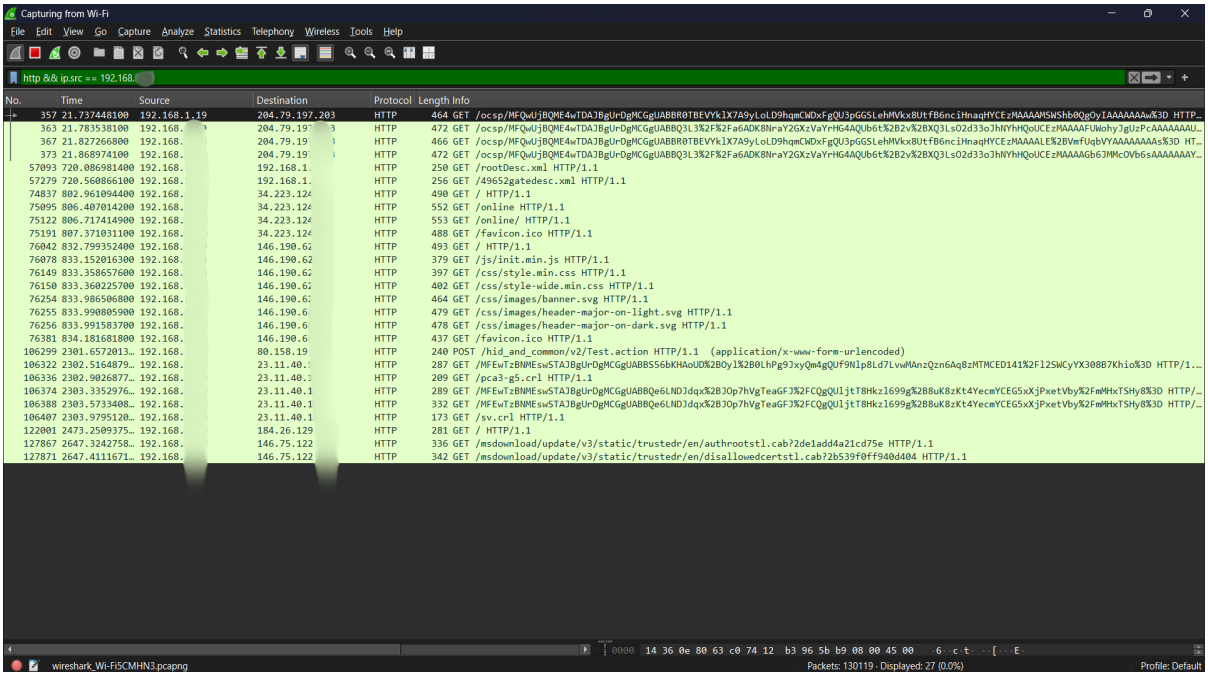

Figure 5: Unencrypted HTTP Traffic

9. Correlation Analysis

The combination of ICMP flooding patterns, suspicious DNS queries, NXDOMAIN responses, and unencrypted HTTP traffic suggests coordinated behavior resembling reconnaissance and potential attack preparation.

10. Threat Assessment

The observed behavior may indicate:

- ICMP Flood or scanning activity

- Domain Generation Algorithm (DGA)

- Command-and-Control (C2) attempts

- Data exposure via HTTP

11. Conclusion

This investigation demonstrates practical SOC skills including traffic analysis, anomaly detection, and threat correlation. The findings highlight the importance of monitoring multi-protocol activity to detect early indicators of compromise.